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Project: End-to-End E-Commerce Analytics Platform

1. Project Overview & Goal

Scenario: You are a data team at a rapidly growing American e-commerce retailer called "**SynthCart**," which specializes in electronics and accessories. The executive team wants to launch a data-driven initiative to understand customer behavior, optimize sales strategies, and improve inventory management.

Goal: To build a robust, end-to-end data platform that ingests raw data from various sources, processes it through a Medallion architecture, and serves it to a business intelligence dashboard for analysis. This project will be a collaborative effort between the Data Engineering (DE) and Data Analysis (DA) teams.

Team Composition:

Data Engineering (DE): 4 members

- Muhammad Awais
- Afnan Khan
- Farheen Muzaffar
- Ghazal E Ashar

Data Analysis (DA): 7 members

- Abdur Rehman
- Aqsa Majeed
- Ayesha Saleh
- Salman Qureshi
- Saud Ijaz
- Wania Nafees
- Zohair Raza

2. Technical Architecture & Tools

This project will utilize the following open-source and free-tier tools.

- **Data Lake:** MinIO (S3-compatible object storage)
- **Data Processing:** Python (Pandas, PySpark) in Google Colab or Jupyter Notebooks
- **Workflow Orchestration:** Apache Airflow
- **Data Warehouse:** PostgreSQL
- **BI & Dashboarding:** Power BI Desktop
- **Version Control & CI/CD:** GitHub
- **Documentation:** MS Word
- **Diagramming:** Draw.io or Lucidchart

3. Dataset: Brazilian E-Commerce Public Dataset by Olist

While the scenario is a US-based company, the **Olist Dataset** is one of the most comprehensive and well-structured e-commerce datasets available. It contains real, anonymized data from a top Brazilian department store and is perfect for this project's scope. It includes information on orders, products, customers, sellers, payments, and reviews.

Data Sources:

1. *Flat Files (Kaggle)*: The core dataset is available on Kaggle and includes multiple CSV files (orders, products, customers, geolocation, etc.). This will simulate batch data ingestion.
 - **Link:** [Olist E-Commerce Dataset on Kaggle](#)
2. *API (Simulated)*: To simulate fetching real-time data, the DE team will use a public API. The **DummyJSON API** provides realistic-looking product and user data that can be creatively joined with the Olist dataset.
 - **Product API Endpoint:** <https://dummyjson.com/products>
 - **User API Endpoint:** <https://dummyjson.com/users>

4. Project Phases & Team Responsibilities

This project follows the **Medallion Architecture (Bronze -> Silver -> Gold)**.

Phase 1: Data Engineering (DE Team Responsibilities)

Your primary goal is to build a reliable and automated data pipeline.

Task 1: Infrastructure & Environment Setup – **Muhammad Awais**

- Set up a local instance of MinIO to act as the data lake.
- Set up a local PostgreSQL database to serve as the data warehouse.
- Set up Apache Airflow (can be run locally via Docker).
- Initialize a GitHub repository for all project code and documentation.

Task 2: Bronze Layer - Raw Data Ingestion – **Afnan Khan**

- Create an Airflow DAG (Directed Acyclic Graph) with two main tasks:
 1. **Batch Ingestion:** A Python script to download the Olist dataset CSV files from Kaggle and load them into a **bronze** bucket in MinIO. The data should be stored in its original raw format.
 2. **API Ingestion:** A Python script to fetch product and user data from the DummyJSON API endpoints and save the raw JSON responses into the bronze bucket in MinIO.
- This DAG should be scheduled to run daily.

Task 3: Silver Layer - Cleaned & Conformed Data – **Farheen Muzaffar**

- Create a second Airflow DAG, triggered upon the successful completion of the Bronze DAG.
- This DAG will run a PySpark job (in a notebook) to perform the following transformations:
 - Read all raw files from the bronze bucket.
 - **Data Cleaning:** Handle missing values, correct data types (e.g., convert date strings to timestamps), and remove duplicates.
 - **Data Validation:** Implement basic data quality checks (e.g., ensure order_id is unique, price is positive).

- **Enrichment:** Join the Olist products data with the product data fetched from the DummyJSON API. You can create a mapping on product_category_name.
- Write the cleaned, transformed, and enriched dataframes to a **silver** bucket in MinIO, preferably in a more optimized format like **Parquet**.

Task 4: Gold Layer - Aggregated Business-Ready Data – Ghazal E Ashar

- Create a third Airflow DAG, triggered upon the successful completion of the Silver DAG.
- This DAG will run a PySpark or SQL job to create aggregated, business-level tables. These are the tables the DA team will use.
- Read the Parquet files from the silver bucket.
- Create the following dimension and fact tables:
 - **dim_customers:** Customer-level details (ID, city, state).
 - **dim_products:** Product-level details (ID, category, price).
 - **dim_sellers:** Seller-level details (ID, city, state).
 - **dim_date:** Date dimension table generated from order timestamps.
 - **fact_orders:** The central fact table containing foreign keys to the dimension tables and key metrics like order_status, payment_value, freight_value, and order_purchase_timestamp.
- Load these final Gold tables from your Spark job into the **PostgreSQL Data Warehouse**.

Deliverables for DE Team:

1. A fully functional Airflow pipeline with three dependent DAGs.
2. All Python/PySpark scripts used for ingestion and transformation, stored in GitHub.
3. An architecture diagram of the data pipeline.
4. Documentation explaining the pipeline, data models, and how to run the project.

Phase 2: Data Analysis (DA Team Responsibilities)

Your primary goal is to extract actionable insights from the data prepared by the DE team.

Task 1: Data Ingestion & Exploration

- Collaborate with DE team and manually load all the Gold layer tables (dim_customers, dim_products, fact_orders, etc.).
- Perform exploratory data analysis (EDA) to understand the data, its distributions, and relationships between tables. Create the necessary table relationships in the Power BI data model view.

Task 2: Dashboard Development

- Create a Power BI report with seven distinct pages. Each team member is responsible for one page. The report should have a consistent theme, branding ("SynthCart"), and interactive filters (e.g., by date, product category, state).

Page 1: Executive Sales Overview – Abdur Rehman

- **KPIs:** Total Revenue, Total Orders, Average Order Value, Sales Over Time (Line Chart).
- **Visuals:** KPI cards, revenue by month/quarter, map visualizing sales by state.

Page 2: Customer Insights – Aqsa Majeed

- **KPIs:** Total Customers, New vs. Returning Customers (if derivable), Customer Acquisition Over Time.
- **Visuals:** Customer distribution by city/state, Top 10 customers by revenue, cohort analysis of customer retention.

Page 3: Product Performance Analysis – Ayesha Saleh

- **KPIs:** Best-Selling Products, Top Performing Product Categories, Average Product Price.
- **Visuals:** Bar chart of revenue by product category, table of top 10 products by units sold, scatter plot of product price vs. review score.

Page 4: Logistics & Fulfillment Analysis – Salman Qureshi

- **KPIs:** Average Delivery Time, On-Time Delivery Rate, Average Freight Value.
- **Visuals:** Chart showing average delivery time by seller state, map of shipping routes, table of orders with delayed delivery.

Page 5: Seller & Review Analysis – Saud Ijaz

- **KPIs:** Number of Active Sellers, Average Seller Rating, Distribution of Review Scores (1 to 5).
- **Visuals:** Bar chart of top 10 sellers by order volume, pie chart of review score distribution, analysis of review comments (e.g., word cloud - advanced goal).

Page 6: Payment Methods Analysis – Wania Nafees

- **KPIs:** Most Common Payment Types, Average Payment Installments, Total Revenue by Payment Method.
- **Visuals:** Bar chart showing the count of each payment_type, KPI card for average payment_installments, pie chart visualizing the share of payment_value by type.

Page 7: Geographic Deep Dive – Zohair Raza

- **KPIs:** Top 10 Cities by Order Volume, Customer vs. Seller Geographic Distribution.
- **Visuals:** A map showing order density by customer city, a dual-axis chart comparing the number of customers and sellers in the top 10 states, analysis of freight value by state.

Deliverables for DA Team:

1. A complete Power BI report file (.pbix).
2. A presentation summarizing the key findings from each dashboard page.
3. Documentation explaining the data model, DAX measures created, and insights.

5. Final Project Deliverables (Combined)

- **GitHub Repository:** A single repository containing all code, documentation, and diagrams from both teams.
- **Final Presentation:** A joint presentation where the DE team presents the data platform they built, and the DA team presents the insights derived from their dashboards. The presentation should tell a cohesive story of how data flows from its raw state to actionable business intelligence.